ANALYSTLAB AFRICA

Experience Lab Internship Programme

Official Weekly Assignment

Week 1: ShopSmart Retail Analytics & AI Implementation Project

Document Control

Programme	AnalystLab Africa Experience Lab
Week	Week 1
Project	ShopSmart Retail Analytics & AI Implementation
Project Title	Delivering a Retail Analytics and Artificial Intelligence Solution for ShopSmart Retail Group
Version	1.0
Issue Date	_____
Submission Deadline	End of Week 1 (Sunday, 11:59 PM WAT)
Submission Method	Official AnalystLab Africa Week 1 Submission Form

1. DATA ANALYTICS TRACK

Project Title

Mission 01: Retail Sales Analysis with Microsoft Excel

Business Scenario

Client

ShopSmart Retail Ltd

ShopSmart Retail Ltd. operates multiple retail stores across Nigeria. The company sells thousands of products across different categories and receives orders from thousands of customers every month.

Management has noticed fluctuations in sales, inventory shortages, and increasing product returns. They have hired **AnalystLab Africa** to analyze the business data and provide actionable recommendations.

You have been assigned as a **Junior Data Analyst**.

Your task is to clean the data, analyze it using Microsoft Excel, create dashboards, and present business insights.

Learning Objectives

By the end of Week 1, interns should be able to:

- Import CSV files into Excel
- Format data professionally
- Clean messy data
- Create Excel Tables
- Use Excel formulas
- Use XLOOKUP/VLOOKUP
- Build Pivot Tables
- Create Pivot Charts
- Build an Executive Dashboard
- Present business insights

Dataset Provided

Interns receive:

Customers.csv
Products.csv
Orders.csv
Stores.csv
Returns.csv
Employees.csv
Suppliers.csv
Inventory.csv
Order_Items.csv
Data_Dictionary.xlsx

PART 1 — Data Understanding (10 Marks)

Open all datasets.

Answer:

1. How many datasets were provided?
2. What does each dataset represent?
3. Identify the Primary Key of each table.
4. Identify possible Foreign Keys.
5. Draw the relationship between the tables.

Deliverable:

A one-page explanation.

PART 2 — Data Cleaning (15 Marks)

Using Excel,

Perform the following:

- Remove duplicate rows
- Check missing values
- Correct inconsistent date formats
- Ensure currency fields are numeric
- Convert datasets into Excel Tables-
- Rename worksheets appropriately

Deliverable:

Cleaned Excel Workbook.

PART 3 — Excel Formulas (20 Marks)

Use formulas to answer the following.

1. Total Revenue
2. Total Quantity Sold
3. Average Selling Price
4. Average Order Value
5. Highest Sales Amount
6. Lowest Sales Amount
7. Number of Orders
8. Number of Customers
9. Number of Products
10. Number of Stores

PART 4 — Lookup Functions (15 Marks)

Using XLOOKUP (or VLOOKUP),

Retrieve:

Customer Name

Store Name

Product Name

Category

Brand

Selling Price

Supplier Name

PART 5 — Pivot Tables (20 Marks)

Create Pivot Tables showing:

Sales by Month

Sales by Store

Sales by Category

Sales by Brand

Sales by State

Sales by Customer Gender

Sales by Membership Level

Top 10 Products

Bottom 10 Products

Top 10 Customers

PART 6 — Charts (10 Marks)

Create the following charts.

Monthly Sales Trend

Revenue by Category

Revenue by Store

Top Products

Sales by Region

Return Rate by Category

PART 7 — Dashboard (30 Marks)

Design an Executive Dashboard containing:

KPI Cards

Total Revenue

Total Orders

Total Customers

Average Order Value

Return Rate

Inventory Value

Charts

Monthly Sales Trend

Revenue by Category

Revenue by Store

Top Products

Sales by State

Interactive Filters

Year

Month

Region

Store

Category

Membership Level

PART 8 — Business Questions (30 Marks)

Answer the following.

1. Which product generated the highest revenue?
2. Which store performed best?
3. Which category sells the most?
4. Which month generated the highest revenue?
5. Which customers spend the most?
6. Which products are frequently returned?
7. Which supplier supplies the highest number of products?
8. Which stores have low inventory?
9. Which region performs best?
10. Which product category should management invest in?

PART 9 — Business Recommendations (20 Marks)

Write a one-page report.

Include:

Three key findings.

Three business recommendations.

One risk identified.

One opportunity identified.

Final Deliverables

Each intern submits:

```
Week1_FirstName_LastName/
├── ShopSmart_Analysis.xlsx
```

```
├── Executive_Dashboard.pdf
├── Business_Report.pdf
└── README.md
```

PART 10 — ASSESSMENT CRITERIA

Section	Marks
Data Understanding	10
Data Cleaning	15
Excel Formulas	20
Lookup Functions	15
Pivot Tables	20
Charts	10
Dashboard	30
Business Questions	30
Business Report	20
Bonus (Creativity & Presentation)	**10**
Total	**180 Marks**

2. DATA SCIENCE TRACK

Week 1 – Introduction to Data Science with Python

Mission DS01: Exploratory Data Analysis (EDA) on Retail Sales Data

Client

ShopSmart Retail Ltd.

ShopSmart Retail Ltd. has accumulated years of customer, product, sales, inventory, and returns data. Before building predictive models, management wants to understand the data and identify hidden patterns.

As a Junior Data Scientist, your responsibility is to perform an Exploratory Data Analysis (EDA) and present insights that will guide future machine learning projects.

Learning Objectives

By the end of Week 1, interns should be able to:

- Understand the Data Science lifecycle
- Load datasets into Python
- Clean data
- Explore data

- Merge multiple datasets
- Calculate descriptive statistics
- Visualize data
- Generate business insights
- Prepare data for machine learning

Dataset

Interns will use the 10 ShopSmart datasets:

Customers.csv
Products.csv
Orders.csv
Stores.csv
Returns.csv
Employees.csv
Suppliers.csv
Inventory.csv
Order_Items.csv
Data_Dictionary.xlsx

Software

- Python 3.x
- Jupyter Notebook
- VS Code (optional)

Required Libraries

```
pandasnumpymatplotlibseabornmissingno
```

Part 1 — Environment Setup (10 Marks)

Tasks

1. Install Python
2. Install Jupyter Notebook
3. Install required libraries
4. Create a notebook named:

```
Week1_EDA.ipynb
```

Part 2 — Load the Data (15 Marks)

Load every CSV into Pandas.

Display:

- Head()
- Tail()
- Shape
- Columns
- Data Types
- Missing Values

Deliverable:

Notebook screenshots and code.

Part 3 — Data Cleaning (20 Marks)

Perform:

- Remove duplicates
- Handle missing values
- Fix incorrect data types
- Convert dates
- Rename columns
- Check inconsistent values

Document every cleaning step.

Part 4 — Exploratory Data Analysis (30 Marks)

Calculate:

- Total Revenue
- Total Customers
- Total Products
- Total Stores
- Total Orders
- Average Order Value
- Average Selling Price
- Total Quantity Sold
- Return Rate
- Inventory Value

Part 5 — Data Visualization (30 Marks)

Create the following charts:

- Monthly Sales Trend
- Revenue by Category
- Revenue by Store
- Revenue by State
- Top 10 Products

- Bottom 10 Products
- Customer Age Distribution
- Product Price Distribution
- Return Rate by Category
- Inventory by Store

Part 6 — Correlation Analysis (20 Marks)

Identify relationships between variables.

Tasks:

- Correlation Matrix
- Heatmap
- Pair Plot (selected numerical features)

Explain:

- Which variables are strongly correlated?
- Which variables appear unrelated?
- How could these relationships be useful?

Part 7 — Feature Engineering (20 Marks)

Create new features such as:

- Profit Margin
- Order Month
- Order Quarter
- Customer Age Group
- Customer Lifetime Value (approximation)
- Discount Percentage
- Days Since Last Purchase
- Inventory Status (Low/Medium/High)

Explain why each feature may improve future models.

Part 8 — Business Questions (30 Marks)

Answer:

1. Which products generate the most revenue?
2. Which stores perform best?
3. Which customer segments spend the most?
4. Which products have high return rates?
5. Which months have peak sales?
6. Which categories are growing?
7. Which suppliers perform best?

8. Which inventory items require restocking?
9. Which regions underperform?
10. What patterns could support predictive modeling?

Support answers with charts or tables.

Part 9 — Data Science Report (25 Marks)

Prepare a professional report containing:

Executive Summary

Dataset Description

Data Cleaning Process

Exploratory Findings

Visualizations

Five Key Insights

Five Recommendations

Suggested Machine Learning Problems

Examples:

- Sales Forecasting
- Customer Churn Prediction
- Product Recommendation
- Return Prediction
- Inventory Forecasting

Part 10 — Code Quality (10 Marks)

Your notebook should include:

- Markdown explanations
- Comments
- Clear variable names
- Modular code
- Well-organized sections

Bonus Challenge (20 Marks)

Choose **one** of the following mini-projects:

1. Predict whether an order will be returned.
2. Segment customers using K-Means clustering.
3. Forecast monthly sales using historical data.
4. Build a product recommendation prototype.
5. Detect unusual transactions using statistical methods.

Document your approach and discuss limitations.

Final Deliverables

```
Week1_DataScience_Project/

├── Week1_EDA.ipynb
├── Cleaned_Data/
├── Visualizations/
├── EDA_Report.pdf
├── requirements.txt
├── README.md
└── Presentation.pptx
```

PART 11 — ASSESSMENT CRITERIA

Section	**Marks**
Environment Setup	10
Data Loading	15
Data Cleaning	20
Exploratory Data Analysis	30
Data Visualization	30
Correlation Analysis	20
Feature Engineering	20
Business Questions	30
Data Science Report	25
Code Quality	10
Bonus Challenge	20
Total	230 **Marks**

Machine Learning Track

Week 1 – Foundations of Machine Learning

Mission ML01: Building Your First Machine Learning Models

Client

ShopSmart Retail Ltd.

ShopSmart Retail Ltd. wants to begin integrating Artificial Intelligence into its retail operations. Before developing advanced AI solutions, the company has hired **AnalystLab Africa** to build its first machine learning prototypes using historical retail data.

As a Machine Learning Engineer, your task is to understand the fundamentals of machine learning, prepare datasets, build baseline models, and evaluate their performance.

Learning Objectives

By the end of Week 1, interns should be able to:

- Understand the Machine Learning lifecycle
- Differentiate supervised, unsupervised, and reinforcement learning
- Understand regression vs. classification
- Prepare data for machine learning
- Build baseline ML models
- Evaluate model performance
- Save trained models

Dataset

Use the **ShopSmart Retail Dataset**.

Customers.csv
Products.csv
Orders.csv
Stores.csv
Returns.csv
Employees.csv
Suppliers.csv
Inventory.csv
Order_Items.csv
Data_Dictionary.xlsx

Software

- Python 3.x
- Jupyter Notebook
- VS Code
- Git

Required Libraries

```
pandasnumpymatplotlibseabornscikit-learnjoblib
```

Part 1 — Machine Learning Fundamentals (15 Marks)

Write a short report explaining:

- What is Machine Learning?
- Types of Machine Learning
- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning
- Regression vs Classification
- Machine Learning Workflow

Part 2 — Data Preparation (20 Marks)

Tasks:

- Load all datasets
- Merge into one master dataset
- Handle missing values
- Remove duplicates
- Encode categorical variables
- Scale numerical variables
- Perform Train/Test split

Deliverable:

```
train.csv
test.csv
```

Part 3 — Build Your First Regression Model (25 Marks)

Business Problem

Predict total order value.

Target Variable

`Order_Total`

Train:

- Linear Regression

Evaluate:

- MAE
- RMSE
- R^2 Score

Visualize:

- Actual vs Predicted
- Residual Plot

Part 4 — Build Your First Classification Model (25 Marks)

Business Problem

Predict whether an order will be returned.

Target Variable

`Return_Flag`

Train:

- Logistic Regression

Evaluate:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix

Part 5 — Understanding Model Evaluation (20 Marks)

Compare regression and classification metrics.

Explain:

- Accuracy
- Precision
- Recall
- F1 Score

- MAE
- RMSE
- R^2

Discuss when each metric should be used.

Part 6 — Feature Importance (15 Marks)

Identify the most influential variables affecting:

- Order Value
- Product Returns

Explain why these features matter from a business perspective.

Part 7 — Business Questions (30 Marks)

Answer:

1. Which variables best predict customer spending?
2. Which variables influence product returns?
3. Which products are easiest to predict?
4. Which customer groups generate the highest order values?
5. Which stores perform best?
6. How could the business use these predictive models?
7. What limitations do the models have?
8. What additional data would improve predictions?

Support your answers with tables and charts.

Part 8 — Model Persistence (15 Marks)

Save the trained models using **Joblib**.

Save:

```
linear_regression.pkl
logistic_regression.pkl
scaler.pkl
encoder.pkl
```

Demonstrate loading the saved models and making predictions.

Part 9 — Documentation (15 Marks)

Prepare a report including:

- Business Problem
- Data Preparation
- Model Development
- Model Evaluation
- Business Insights
- Recommendations
- Limitations

Bonus Challenge (20 Marks)

Create a simple **prediction script** that:

- Loads the saved model
- Accepts user input
- Predicts:
  - Order Value
  - Return Probability
- Displays the prediction in the terminal

Final Deliverables

```
Week1_MachineLearning/

├── ML01_Foundations.ipynb
├── data/
│   ├── train.csv
│   ├── test.csv
│   └── master_dataset.csv
├── models/
│   ├── linear_regression.pkl
│   ├── logistic_regression.pkl
│   ├── scaler.pkl
│   └── encoder.pkl
├── scripts/
│   └── predict.py
├── reports/
│   ├── ML_Report.pdf
│   └── Presentation.pptx
├── requirements.txt
├── README.md
└── .gitignore
```

PART 10 — ASSESSMENT CRITERIA

Section	Marks
Machine Learning Fundamentals	15
Data Preparation	20
Regression Model	25

Classification Model	25
Model Evaluation	20
Feature Importance	15
Business Questions	30
Model Persistence	15
Documentation	15
Bonus Challenge	20
Total	**200 Marks**

Generative AI Engineering Track

Week 1 – AI Retail Copilot with Prompt Engineering & Foundation Models

Mission GEN01: Build the ShopSmart AI Retail Copilot

Client

ShopSmart Retail Group

ShopSmart Retail Group is one of Africa's fastest-growing supermarket chains, operating multiple stores across different cities. The company generates thousands of transactions every day and stores data on customers, products, orders, inventory, suppliers, employees, returns, and store operations.

Although the company has already built dashboards and predictive models through its Data Analytics, Data Science, and Machine Learning teams, business managers still struggle to extract actionable insights because they cannot easily query the data or interpret analytical results.

ShopSmart wants to build an **AI Retail Copilot** that allows executives, store managers, inventory officers, sales teams, and customer service representatives to interact with retail data using natural language.

AnalystLab Africa has been contracted to design and build the first version of the ShopSmart AI Retail Copilot.

As a **Generative AI Engineer**, your responsibility is to build an AI assistant capable of understanding retail datasets, generating business insights, summarizing reports, assisting with inventory decisions, and answering operational questions.

Learning Objectives

By the end of Week 1, interns should be able to:

- Understand Generative AI fundamentals
- Apply Prompt Engineering techniques

- Connect LLMs to structured retail data
- Build an AI-powered retail assistant
- Generate business reports using AI
- Create intelligent business prompts
- Evaluate LLM performance in retail use cases

Business Dataset

Interns must use the **ShopSmart Week 1 Retail Dataset** created in the Data Analytics track.

Required datasets:

Customers.csv
Products.csv
Orders.csv
Stores.csv
Returns.csv
Employees.csv
Suppliers.csv
Inventory.csv
Order_Items.csv
Data_Dictionary.xlsx

Interns are encouraged to also use outputs from:

- Data Analytics dashboards
- Data Science notebooks
- Machine Learning sales prediction model (if completed)

Part 1 — Understanding Generative AI in Retail (15 Marks)

Prepare a report explaining:

- Artificial Intelligence
- Generative AI
- Large Language Models
- Prompt Engineering
- Tokens
- Context Windows
- Embeddings
- Transformers
- Responsible AI

Explain how Generative AI can improve:

- Retail operations
- Inventory management
- Customer service
- Executive decision-making

Part 2 — Retail Foundation Model Comparison (20 Marks)

Compare at least five LLMs for retail applications.

Include:

- GPT-5 / GPT-4o
- Gemini
- Claude
- Llama
- Mistral

Evaluate:

- Business reasoning
- Data summarization
- Table understanding
- Cost
- Speed
- Context Window

Recommend the best model for ShopSmart.

Part 3 — Retail Prompt Engineering (35 Marks)

Design prompts for ShopSmart managers.

Example prompts:

Executive

Summarize today's retail performance.

Inventory

Which products are running low?

Sales

Which categories generated the highest revenue?

Supplier

Which supplier has delayed deliveries?

Customer

Which customers made the highest purchases?

HR

Which employees generated the most sales?

Store Manager

Compare Store A and Store B.

Finance

Generate this week's executive sales report.

Evaluate:

- Zero-shot
- One-shot
- Few-shot
- Role prompting
- Chain-of-thought
- Structured prompting

Compare quality.

Part 4 — AI Business Report Generation (30 Marks)

Use an LLM to generate:

- Daily Sales Report
- Weekly Executive Report
- Monthly Inventory Summary
- Customer Insights Report
- Supplier Performance Report
- Product Performance Report
- Store Performance Report
- Return Analysis Report

Evaluate the quality and usefulness of each report.

Part 5 — AI Retail Decision Support (35 Marks)

Build prompts that assist business decision-making.

Example tasks:

- Recommend products for promotion.
- Identify products to reorder.
- Suggest inventory transfers between stores.
- Recommend pricing strategies.

- Detect unusual sales patterns.
- Explain reasons for declining sales.
- Recommend cross-selling opportunities.
- Suggest actions to reduce product returns.

Document how AI supports managerial decisions.

Part 6 — AI Retail Copilot Development (40 Marks)

Develop a **Streamlit** application.

Features:

- Retail chatbot
- Upload CSV datasets
- Ask business questions
- Dataset summary
- Executive dashboard summary
- Conversation history
- Export responses
- System prompt customization

Part 7 — Retail Prompt Evaluation (20 Marks)

Evaluate prompts based on:

- Accuracy
- Business relevance
- Completeness
- Explainability
- Hallucination rate
- Response consistency

Recommend the best prompting strategy for retail analytics.

Part 8 — Business Questions (30 Marks)

Answer:

1. How can Generative AI improve retail operations?
2. Which foundation model is best suited for ShopSmart?
3. Which prompting strategy generated the most useful business insights?
4. How can AI assist store managers?
5. How can AI improve inventory management?
6. How can AI support customer service teams?
7. What are the limitations of using LLMs with structured retail data?
8. What ethical considerations should ShopSmart address before deploying an AI assistant?
9. How can AI complement traditional analytics dashboards?

10. What future enhancements would you recommend for the ShopSmart AI Retail Copilot?

Support your answers with examples and outputs from your application.

Part 9 — Documentation (20 Marks)

Prepare documentation including:

- Project Overview
- Prompt Design
- Model Comparison
- AI Retail Copilot Architecture
- Dataset Description
- Evaluation Results
- Business Recommendations
- Ethical Considerations

Bonus Challenge (20 Marks)

Enhance the AI Retail Copilot with one or more of the following:

- Chat with multiple CSV files simultaneously.
- Generate charts from user questions.
- Integrate the Week 1 sales prediction model from the Machine Learning track.
- Support voice-based questions.
- Add multilingual support (e.g., English and French).
- Generate PowerPoint-ready executive summaries.

Final Deliverables

```
Week1_GenerativeAI_Retail_Copilot/

├── notebooks/
│   └── GEN01_Retail_Copilot.ipynb
├── data/
│   ├── ShopSmart_Customers.csv
│   ├── ShopSmart_Products.csv
│   ├── ShopSmart_Orders.csv
│   ├── ShopSmart_Order_Items.csv
│   ├── ShopSmart_Stores.csv
│   ├── ShopSmart_Employees.csv
│   ├── ShopSmart_Suppliers.csv
│   ├── ShopSmart_Inventory.csv
│   ├── ShopSmart_Returns.csv
│   └── ShopSmart_Data_Dictionary.xlsx
├── prompts/
│   ├── executive_prompts.md
│   ├── inventory_prompts.md
```

```
│   ├── sales_prompts.md
│   ├── finance_prompts.md
│   └── customer_service_prompts.md
├── app/
│   └── streamlit_app.py
├── reports/
│   ├── Retail_AI_Report.pdf
│   ├── Prompt_Evaluation_Report.pdf
│   └── Executive_Presentation.pptx
├── screenshots/
├── requirements.txt
├── README.md
└── LICENSE
```

PART 10 — ASSESSMENT CRITERIA

Section	Marks
Generative AI Fundamentals	15
Foundation Model Comparison	20
Retail Prompt Engineering	35
AI Business Report Generation	30
AI Retail Decision Support	35
AI Retail Copilot Development	40
Prompt Evaluation	20
Business Questions	30
Documentation	20
Bonus Challenge	20
Total	**265 Marks**

Project Management Track

Week 1 – Project Initiation & Planning

Mission PM01: Manage the ShopSmart Retail Analytics & AI Implementation Project

Client

ShopSmart Retail Group

ShopSmart Retail Group is a leading retail chain with multiple stores across Nigeria. The company wants to improve business performance by transforming its retail data into actionable insights and intelligent business solutions.

To achieve this goal, ShopSmart has contracted **AnalystLab Africa Consulting** to deliver a Retail Analytics and Artificial Intelligence solution.

The engagement consists of four technical workstreams:

- Data Analytics
- Data Science
- Machine Learning
- Generative AI

As the **Project Manager**, your responsibility is to successfully initiate, plan, coordinate, monitor, and close the ShopSmart Retail Analytics & AI Implementation Project while ensuring that all project objectives are delivered within the agreed scope, schedule, quality, and budget.

Business Problem

ShopSmart currently manages data across multiple systems but lacks an integrated solution for business intelligence and AI-driven decision-making.

The company needs a project team to deliver:

- Interactive sales dashboards
- Retail data analysis
- Sales prediction models
- AI Retail Copilot

using the ShopSmart retail datasets.

Project Objectives

By the end of this project, ShopSmart should have:

- Executive Sales Dashboard
- Customer Analytics Dashboard
- Inventory Performance Dashboard
- Clean Retail Dataset
- Feature Engineering Pipeline
- Sales Forecasting Model
- AI Retail Copilot
- Final Business Presentation
- Project Documentation

Project Scope

In Scope

The project includes planning and coordinating the delivery of:

Workstream 1 – Data Analytics

Deliverables

- Sales Dashboard
- Customer Dashboard
- Store Performance Dashboard
- Inventory Dashboard
- Executive Report

Workstream 2 – Data Science

Deliverables

- Data Cleaning
- Data Validation
- Exploratory Data Analysis
- Feature Engineering
- Statistical Insights

Workstream 3 – Machine Learning

Deliverables

- Sales Prediction Model
- Customer Segmentation
- Product Recommendation Prototype
- Model Evaluation Report

Workstream 4 – Generative AI

Deliverables

- AI Retail Copilot
- Retail Report Generator
- Prompt Library
- Business Question Answering Assistant

Out of Scope

- ERP implementation
- Mobile application development
- Live deployment to production
- Payment gateway integration
- Customer loyalty system

Learning Objectives

By the end of Week 1, interns should be able to:

- Understand project management fundamentals
- Develop a Project Charter
- Define project scope
- Identify project stakeholders
- Create a Work Breakdown Structure (WBS)
- Develop a project schedule
- Allocate project resources
- Perform risk analysis
- Prepare communication plans
- Coordinate multidisciplinary project teams

Project Dataset

The Project Manager should understand the business assets being managed.

Project datasets include:

Customers.csv
Products.csv
Orders.csv
Stores.csv
Returns.csv
Employees.csv
Suppliers.csv
Inventory.csv
Order_Items.csv
Data_Dictionary.xlsx

Part 1 — Develop the Project Charter (20 Marks)

Prepare a Project Charter containing:

- Project Name
- Business Case
- Problem Statement
- Objectives
- Deliverables
- Scope
- Assumptions
- Constraints
- Success Criteria
- Project Sponsor
- Project Manager
- Approval Section

Part 2 — Stakeholder Identification & Analysis (20 Marks)

Identify all project stakeholders.

Examples:

- CEO
- COO
- Store Managers
- Inventory Manager
- Sales Manager
- Finance Manager
- IT Manager
- AnalystLab Consulting Team

Develop:

- Stakeholder Register
- Stakeholder Analysis Matrix
- Power–Interest Grid
- Stakeholder Engagement Plan

Part 3 — Scope Management (20 Marks)

Prepare:

- Scope Statement
- In-Scope Deliverables
- Out-of-Scope Items
- Scope Baseline
- Scope Verification Plan
- Scope Change Process

Part 4 — Work Breakdown Structure (30 Marks)

Develop a complete WBS for the ShopSmart project.

Include work packages for:

- Project Management
- Data Analytics
- Data Science
- Machine Learning
- Generative AI
- Testing
- Documentation
- Final Presentation
- Project Closure

Assign responsibilities for each work package.

Part 5 — Project Schedule (25 Marks)

Create:

- Activity List
- Task Dependencies
- Milestones
- Gantt Chart
- Timeline
- Critical Path (optional)

Part 6 — Resource Planning (20 Marks)

Prepare a Resource Management Plan.

Include:

Human Resources

- Project Manager
- Business Analyst
- Data Analyst
- Data Scientist
- Machine Learning Engineer
- Generative AI Engineer
- QA Tester

Technology Resources

- Power BI
- Python
- Jupyter Notebook
- VS Code
- GitHub
- Streamlit
- OpenAI/Gemini API

Part 7 — Risk Management (25 Marks)

Develop a Risk Register.

Potential risks:

- Poor data quality
- Missing records
- Delayed deliverables
- Scope creep
- API limitations

- Budget constraints
- Communication breakdown
- Technical integration issues

For each risk include:

- Description
- Probability
- Impact
- Risk Score
- Mitigation Strategy
- Contingency Plan
- Risk Owner

Part 8 — Communication Management (20 Marks)

Prepare:

- Communication Plan
- Weekly Status Report Template
- Meeting Agenda Template
- Escalation Matrix
- Project Status Dashboard

Part 9 — Quality Management (20 Marks)

Develop a Quality Management Plan covering:

- Deliverable acceptance criteria
- Dashboard quality standards
- Code review process
- Testing approach
- Documentation standards
- Final acceptance checklist

Part 10 — Business Questions (20 Marks)

Answer the following:

1. Why is project management critical to analytics and AI projects?
2. Which project management methodology would you recommend (Agile, Scrum, Hybrid, or Waterfall) and why?
3. What are the biggest risks facing the ShopSmart project?
4. How should changes to project scope be managed?
5. How can effective stakeholder communication improve project success?
6. What KPIs should the Project Manager monitor throughout the project?
7. How will you measure project success?
8. What lessons from this project could be applied to future AI implementations?

Support your answers with examples from the ShopSmart project.

Part 11 — Documentation (20 Marks)

Prepare a complete Project Initiation Package including:

- Executive Summary
- Project Charter
- Stakeholder Register
- Scope Statement
- WBS
- Project Schedule
- Gantt Chart
- Resource Plan
- Risk Register
- Communication Plan
- Quality Management Plan
- Project Management Report

Bonus Challenge (20 Marks)

Develop any **two** of the following:

- RACI Matrix
- RAID Log
- Responsibility Assignment Matrix
- Budget Estimation Sheet
- Change Request Form
- Issue Log
- Lessons Learned Register
- Project Dashboard (Power BI or Excel)

Final Deliverables

```
Week1_Project_Management_ShopSmart/

├── 01_Project_Charter.docx
├── 02_Stakeholder_Register.xlsx
├── 03_Power_Interest_Grid.xlsx
├── 04_Scope_Statement.docx
├── 05_Work_Breakdown_Structure.xlsx
├── 06_Project_Schedule.xlsx
├── 07_Gantt_Chart.xlsx
├── 08_Resource_Plan.xlsx
├── 09_Risk_Register.xlsx
├── 10_Communication_Plan.docx
├── 11_Quality_Management_Plan.docx
├── 12_Project_Management_Report.pdf
├── 13_Executive_Presentation.pptx
├── README.md
└── Project_Repository_Link.txt
```

Part 12 - ASSESSMENT CRITERIA

Section	Marks
Project Charter	20
Stakeholder Analysis	20
Scope Management	20
Work Breakdown Structure	30
Project Schedule	25
Resource Planning	20
Risk Management	25
Communication Management	20
Quality Management	20
Business Questions	20
Documentation	20
Bonus Challenge	20
Total	**260 Marks**

Professional Development Requirement

As part of building your professional portfolio:

- Update your GitHub repository with your Week 1 project.
- Publish a LinkedIn post describing your project
- Publish an X (Twitter) post summarizing your project.
- Tag AnalystLab Africa on LinkedIn.
- Tag @analystlabafric on X.
- Include the hashtag #AnalystLabAfrica on both platforms.

Professional Branding Reminder

AnalystLab Africa encourages every intern to document their learning journey professionally on LinkedIn and X. Consistently sharing your work helps build your portfolio and increases your visibility to recruiters and employers.